

AI-Driven Bond Pricing

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Today's agenda



Problem Statement



Milestones



Problem Formulation



Plots



Evaluation Results



Conclusion

What are bonds?

- ❑ Bonds are a type of loan that is given out from an investor to an entity (government bonds, corporate bonds, municipal bonds) for a set period of time, with regular interest payments and the return of the principal (total sum) on the maturity date (pay date).
- ❑ The bond market includes Illiquid and liquid bonds.
 - ❑ Illiquid bonds are harder to sell due to their uniqueness (fewer buyers and sellers in the market).
 - ❑ We cannot compare the model's price to a single correct market price since it doesn't exist. Mean absolute error (MAE) would be insufficient for this case.

So what are we proposing?

Research Question: How can we develop an appropriate machine learning model that can properly assess the reliability of AI-driven bond pricing models for liquid bonds?



Milestones

Today's agenda



Milestones

1. Implementation of an API for data collection

2. Data collection for Liquid bonds

Format into a CSV file to read from

3. Gradient Boosting analysis as a model and its applications

4. Apply modeling (Linear/Ridge regression)

5. Feature engineering

6. Calculate errors/loss function (Mean Absolute Error, R2 Error)

7. Evaluate and interpret results

Are they accurate for market pricing?

8. Fit data to plots

How do we tackle our research question?

These are some of the key characteristics that we wish to include in our creation of our model:

- ☐ We want to evaluate our model on trained liquid bonds that are traded, with different time-ranges based on their test performance.
- ☐ We want smoothness in the market to reflect and showcase stability.
- ☐ We want to consider the fact that a bond with a longer maturity should have a higher yield.
- ☐ Accurate pricing of bonds in our test data.

What are some of our features?

We'll need to consider the following variables:

- ☐ Bond characteristics
 - ☐ Coupon rate
 - ☐ Maturity date
 - ☐ Issue amount
 - ☐ Price
 - ☐ Updating date
- ☐ Time-Series Features

Model Training and Architecture

We are choosing to utilize and test two different models:

- ❑ Ridge Regression

$$\min_{\beta} \{ \sum_{i=1}^n (y_i - X_i \beta)^2 + \lambda \sum_{j=1}^p \beta_j^2 \}$$

- ❑ Gradient Boosting

$$F^* = \operatorname{argmin}_{F_{x,y}} E [L(y, F(x))]$$

With the following evaluation metrics:

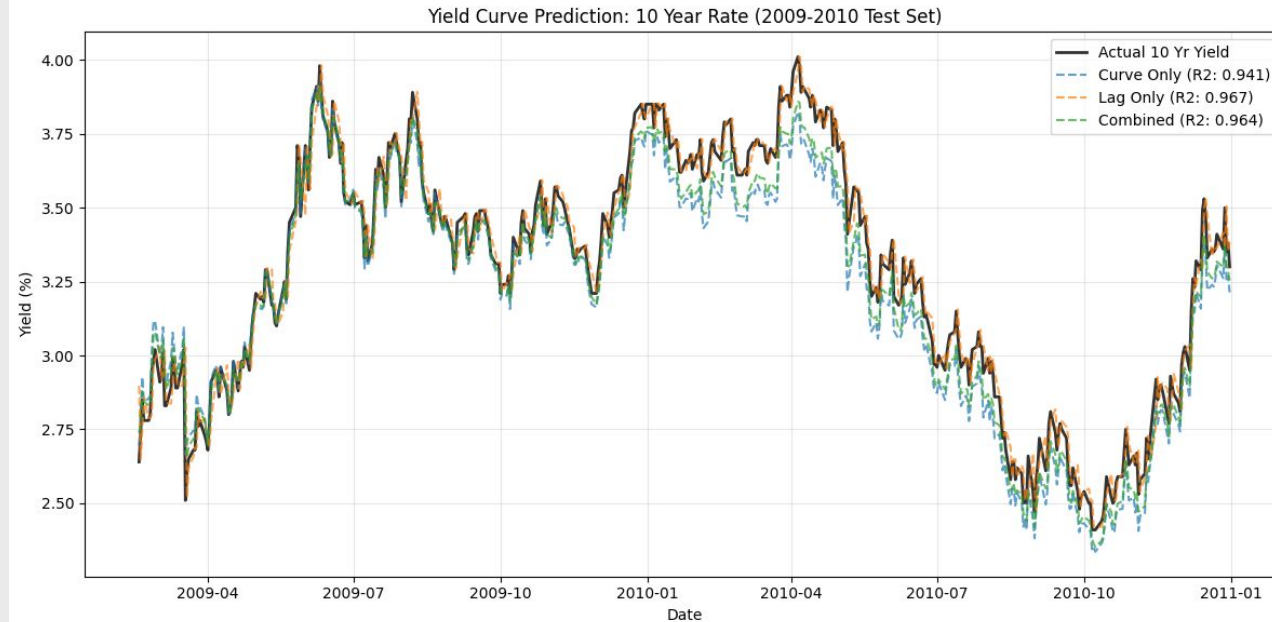
- ❑ Mean Absolute Error

$$MAE = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i|$$

- ❑ R^2 Score

$$R^2 = 1 - \frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{\sum_{i=1}^n (y_i - \bar{y})^2}$$

Yield Curve Prediction



R² and MAE Metrics

With our dataset, we compared the yield curve, lagged yield, and combined (both lagged and curved).

The graph showcases the high accuracy from our lag only model, with an R2 score of 0.967.

Results from Data Plots



Evaluation - Model Performance

1

Curve Only

The "curve only" model showcased an R2 score of 0.941 and an MAE score of 0.085

2

Lag Only

The "lag only" model has an R2 score of 0.966 and an MAE score of 0.057.

3

Combined

The "combined" model has an R2 score of 0.964 and an MAE of 0.067.

4

Best Performing Model

The Lag Only model performs best, with the highest R2 score out of the three.



Thank you

If you have any questions, contact:
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